Back-of-the-Envelope Estimation (HLD)

What is Back-of-the-Envelope Estimation?

Back-of-the-Envelope Estimation is a quick approximation technique used during System Design interviews to estimate:

- System traffic and throughput
- Storage requirements
- Memory/cache requirements
- Number of servers
- Network bandwidth
- Potential bottlenecks

The goal is not exact accuracy but obtaining the correct order of magnitude.

Common Conversions

Traffic & Storage Cheat Sheet

Number	Traffic	Storage
10^3	Thousand	KB
10^6	Million	MB
10^9	Billion	GB
10^{12}	Trillion	TB
10^{15}	Quadrillion	PB

Approximate Data Sizes

Data Type	Size
Character	2 Bytes
Integer	4 Bytes
Long / Double	8 Bytes
Small Text Post (250 chars)	500 Bytes

Data Type	Size
Image	300 KB
HD Video (1 min)	5-10 MB

Example: Facebook Estimation

Assumptions

User Base

- Total Registered Users = 1 Billion
- Daily Active Users (DAU) = 25%

```
DAU = 1B × 25%
    = 250 Million Users
```

User Activity

Each active user performs:

```
5 Reads/day
2 Writes/day
```

Total:

```
7 Requests/User/Day
```

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1. Traffic Estimation

Total requests/day:

```
250M × 7
= 1.75 Billion Requests/Day
```

Requests per second (QPS):

```
1.75B / 86400
≈ 20,250 Requests/sec
≈ 20K QPS
```

Peak Traffic

A common rule:

```
Peak QPS ≈ 2 × Average QPS
```

```
Peak QPS ≈ 40K QPS
```

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2. Storage Estimation

Text Posts

Assume:

```
2 posts/user/day
250 characters/post
```

Storage per post:

```
250 × 2 Bytes
= 500 Bytes
```

Per user:

```
2 × 500
= 1000 Bytes
≈ 1 KB
```

Daily storage:

```
250M × 1 KB
= 250 GB/day
```

Yearly storage:

```
250 × 365
≈ 91 TB/year
```

5-year storage:

```
≈ 455 TB
≈ 500 TB
```

Image Storage

Assume:

```
10% of DAU upload one image/day
```

```
25M Images/Day
```

Image size:

```
300 KB
```

Daily image storage:

```
25M × 300 KB

= 7,500,000 MB
= 7,500 GB
≈ 7.5 TB/day
```

Yearly:

```
7.5 × 365
≈ 2.7 PB/year
```

5 years:

```
≈ 13.5 PB
```

Total Storage

Text Posts

```
≈ 500 TB (5 years)
```

Images

```
≈ 13.5 PB (5 years)
```

Storage is dominated by images.

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3. Cache / Memory Estimation

Assume we cache:

```
Last 5 posts per user
```

Storage per user:

```
5 × 500 Bytes
= 2500 Bytes
≈ 2.5 KB
```

Total cache:

```
250M × 2.5 KB

= 625 GB
```

Round up:

```
≈ 700 GB Cache
```

Including replicas and overhead:

```
≈ 1 TB RAM
```

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4. Number of Cache Servers

Assume:

```
Usable RAM per server = 75 GB
```

Servers required:

```
1000 / 75
≈ 14 Servers
```

Round up:

```
15 Cache Servers
```

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5. Application Server Estimation

Assume:

```
Average latency = 500 ms
```

One thread can serve:

```
1 / 0.5
= 2 Requests/sec
```

If a server has:

```
50 Worker Threads
```

Then:

```
50 × 2
= 100 Requests/sec
```

Required QPS:

```
20,000 Requests/sec
```

Servers needed:

```
20,000 / 100
= 200 Servers
```

For redundancy:

```
250–300 Application Servers
```

Final Estimation Summary

Metric	Value
DAU	250 Million
Average QPS	20K
Peak QPS	40K
Text Storage (5 years)	~500 TB
Image Storage (5 years)	~13.5 PB

Metric	Value
Cache Requirement	~1 TB
Cache Servers	~15
Application Servers	~200-300

Important Interview Tips

1. Always calculate:

2. QPS

3. Storage
4. Cache

5. Server Count

6. Mention Peak Traffic:

```
Peak QPS = 2–5 × Average QPS
```

1. Images and videos dominate storage.

2. Cache only hot data (10–20% of frequently accessed data).

3. Round values aggressively. Precision is less important than order-of-magnitude correctness.

CAP Theorem Tradeoff for Facebook

For most Facebook features:

```
Availability + Partition Tolerance (AP)
```

Reason:

- Users prefer seeing slightly stale data rather than service downtime.
- News Feed, Likes, Comments, Followers can tolerate eventual consistency.
- System must remain available during network partitions.

Some critical services may use stronger consistency internally, but the overall platform leans toward AP with eventual consistency.